



**EEHC DISTRIBUTION MATERIALS SPECIFICATION**

**EDMS 17-401-2**

**Date: 29-09-2020**

**EDMS 17-401-2**

**TECHNICAL SPECIFICATION**

**FOR**

**M.V. VOLTAGE TRANSFORMERS**



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Issue: SEP-2020 / Rev- 2

❖ توجد بنود اختيارية Option items يجب تحديدها بواسطة شركة التوزيع قبل الطرح.



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**CONTENTS**

|                                  |   |
|----------------------------------|---|
| 1. Scope.....                    | 3 |
| 2. General description.....      | 3 |
| 3. Applicable Standards.....     | 3 |
| 4. Environmental conditions..... | 3 |
| 5. Technical specification.....  | 4 |
| 6. Design criteria.....          | 4 |
| 7. Type test.....                | 5 |
| 8. Routine test.....             | 5 |
| 9. Inspection.....               | 5 |
| 10. Marking.....                 | 5 |
| 11. Drawings and Catalogues..... | 5 |
| 12. Sample.....                  | 6 |
| 13. Guarantee Table.....         | 6 |
| 14. Warranty Period.....         | 6 |
| 15. Notes.....                   | 6 |
| 16. Guarantee table.....         | 7 |



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EDMS 17-401-2

Date: 29-09-2020

### Scope:

This specification covers the minimum technical requirements for design, manufacturing, testing, packing and procurement MV voltage transformer used for measuring and protective relays at 50HZ.

### General description:

Single-pole insulated voltage transformers casted in epoxy resin and designed for insulation voltages of 12 kV and 24 KV. The transformer is equipped with two secondary windings (can be three if required), the first one for measuring and the other for protection. The transformer can be mounted in any position. The secondary terminal board should be covered with a transparent and sealable cover made of suitable insulated material.

In case of Voltage transformer equipped with fuse, the fuse link should be easy to be replaced (According to distribution company requirement).

Unless otherwise specified, the transformers are manufactured conformably to the requirements and recommendations of IEC.

### Applicable standards:

Unless otherwise specified, the voltage transformer should comply with the latest edition of IEC publications or any other equivalent international standard specification approved by (....EDC).

- . IEC 61869-1, General requirements
- . IEC 61869-3, Additional requirements for inductive voltage transformers
- . IEC 60282-1, high voltage fuses
- . And any relevant IEC standards

### Environmental conditions:

The performance of the voltage transformer should be guaranteed for the following table, any differences in the guaranteed performance should be clearly set out in the offer:

|                             |                         |
|-----------------------------|-------------------------|
| Minimum ambient temperature | -5°c                    |
| Maximum ambient temperature | 45°c (50 °c as option). |
| Maximum relative humidity   | 95%.                    |
| Maximum altitude            | 1000 m.                 |



## EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 17-401-2

Date: 29-09-2020

### Technical specification:

| Items                               | Value                                                     |
|-------------------------------------|-----------------------------------------------------------|
| Rated Primary Voltage (V)           | 11000/ $\sqrt{3}$ – 10500/ $\sqrt{3}$ – 22000/ $\sqrt{3}$ |
| Rated Secondary Voltage (V)         |                                                           |
| 1 <sup>st</sup> Winding             | 110/ $\sqrt{3}$ – 100 / $\sqrt{3}$ – 220/ $\sqrt{3}$      |
| 2 <sup>nd</sup> Winding             | 110/3 - 100/3 – 220/3                                     |
| 3 <sup>rd</sup> Winding             | As required                                               |
| Rated Burden                        |                                                           |
| 1 <sup>st</sup> winding             | 100 VA                                                    |
| 2 <sup>nd</sup> winding             | 100 VA                                                    |
| Accuracy class                      |                                                           |
| 1 <sup>st</sup> winding             | 0.5                                                       |
| 2 <sup>nd</sup> winding             | 3P                                                        |
| 3 <sup>rd</sup> winding as required | As required                                               |

### Design criteria:

- Rated frequency 50 HZ.
- Creepage distance  $\geq 2$  Cm/KV
- Neutral earthed through low resistance  $< 5$  Ohm
- Voltage factor 1.2 continuous – 1.9 for 8 hours
- The transformer is equipped with two secondary windings, the first one for measuring and the other for protection.
- Rated impulse withstands voltage for primary windings 75 KV for 11KV and 125 KV for 22 KV.
- Short duration withstand voltage (r.m.s) for primary windings (1min.) 28 KV for 12 KV and 50KV for 22 KV.
- Rated power frequency short duration withstand voltage for secondary winding (1 min.) 3 KV.



## EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 17-401-2

Date: 29-09-2020

- Maximum winding temperature rise should be according to IEC 61869-1, 3.
- Limits of voltage error and phase displacement should be in accordance with IEC 61869-1, 3.
- Minimum class of insulation is E temperature rise limit "K" 70 °C @ 45°C ambient (65@50°C ambient) as shown at Table 5 in IEC 61869-1.
- The V.T. Transformation ratio should be as per above table
- The rated burden should be as per above table.

### **Type tests:**

- Temperature-rise test
- Impulse voltage withstand test on primary terminals
- Tests for accuracy
- Short-time current tests

Type tests should be in accordance with IEC 61869-1,3.

Type test certificates should be submitted with the offer.

### **Routine/acceptance tests: -**

1. Power-frequency voltage withstand tests on primary terminals
2. Partial discharge measurement.
3. Power-frequency voltage withstand tests between sections
4. Power-frequency voltage withstand tests on secondary terminals.
5. Test for accuracy.
6. Verification of markings.

Acceptance tests should be performed on random samples and should be selected and witnessed by Distribution Company representative.

Routine tests should be performed in compliance with IEC61869-1,3

### **Inspection:**

- 1) ..... Electricity Distribution Company (.....EDC) preserves its right to witness acceptance tests of voltage transformer.
- 2) Unless otherwise specified or approved in writing by (.....EDC) all voltage transformers should be tested in accordance with all relevant latest applicable IEC standards and test report certificate should be provided with it. (.....EDC) may carryout testing at their lab. The VT will be assumed as rejected if any functions of the VT/s are found faulty.

### **Marking: -**

The following data should be marked on VT itself or on a nameplate engraved or laser printed on the VT, and located in a place such that they are visible and legible when VT is installed:



## EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 17-401-2

Date: 29-09-2020

- Manufacturer name
- Country of origin
- VT type
- Serial number and order number
- Year of manufacture
- No of IEC standard
- Rated voltage (Vn)
- Rated frequency.

All V.T. should carry at least the clear and indelible markings as specified in IEC 61869-1, 3.

### **Drawings and Catalogues: -**

The tenderer should submit a complete set of dimensional drawings, technical catalogues and user manual showing the VT type offered and its performance in full details.

### **Sample: -**

The tenderer should provide a sample V.T. with his offer (if Distribution Company required). The sample should correspond in all respects with the material they intend to supply.

### **Guarantee Table:**

Tenders not accompanied with attached table of guarantee fulfilled and signed by tenderers should not be considered.

### **Warranty Period:**

Warranty period should be 18 months from delivery or 12 months from operation date which later. For the first submittal, the warranty period should be extended to be 3 years at least ended with a performance report prepared by corresponding...EDC. This report should prove product high quality.

### **ملاحظة:**

يلزم تقديم (ضمن مستندات العرض) نسخة من شهادة اختبار من مركز أبحاث الجهد الفائق تفيد اجتياز محولات الجهد من نفس الطراز ولنفس الجهد المقنن للاختبارات النوعية التي أجريت عليها و تستكمل باقي الاختبارات النوعية طبقا للمواصفات القياسية IEC علي ان يتم الاطلاع علي أصل أو صورة طبق الأصل لشهادات الاختبارات النوعية.



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**GUARANTEE TABLE**

|                                                                    |    |            |
|--------------------------------------------------------------------|----|------------|
| - Standard/s                                                       |    |            |
| - Name of Manufacturer                                             |    |            |
| - Place of manufacturing                                           |    |            |
| - Year of manufacturing                                            |    |            |
| - Product type                                                     |    |            |
| - Highest voltage for equipment $U_m$ (r.m.s)                      | KV |            |
| - Rated primary service voltage                                    | KV |            |
| - Rated frequency                                                  | Hz |            |
| - Rated primary voltage                                            | V  |            |
| - Rated secondary voltage                                          | V  |            |
| - Winding 1                                                        |    |            |
| - Winding 2                                                        |    |            |
| - Winding 3 (if required)                                          |    |            |
| - Voltage factor                                                   |    |            |
| - Ambient max. temperature                                         | °C |            |
| - Rated impulse withstand voltage for primary windings             | KV |            |
| - Short duration withstand (r.m.s) voltage for primary windings    | KV |            |
| - Rated power frequency short duration withstand for secondary     | KV |            |
| - Rated Burden                                                     | VA |            |
| - Winding 1                                                        |    |            |
| - Winding 2                                                        |    |            |
| - Winding 3 (if required)                                          |    |            |
| - Accuracy class                                                   |    |            |
| - Winding 1                                                        |    |            |
| - Winding 2                                                        |    |            |
| - Winding 3 (if required)                                          |    |            |
| - Creepage distance                                                | Cm |            |
| - Class of insulation                                              |    |            |
| - Marking:                                                         |    |            |
| . Marking of the rating plate should be according to IEC 61869-1-3 |    | (YES / NO) |

I/We certify that the above mentioned data is guaranteed for the offered cable

Date: / /

Signature